

Lec 44 Introduction to Diffusion Model

Lec 44 Introduction to Diffusion Model

Lecture	44 / 63 + intro
Duration	37:23
Captions	English (auto-generated)
Words	5,226
Video	scQsAvXIOWw
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In today's session, we will discuss about diffusion probabilistic models. Okay? So, the contents for today's session goes like evolution from autoencoders to diffusion models, introduction to diffusion model, need for diffusion model, forward and reverse diffusion process, training and inference pipeline, and a brief introduction to Markov chain.

[00:36] So, we will start with the evolution from autoencoders to diffusion models. In my previous session, we have studied about StyleGAN 2. And before that one, we have studied about normal GANs. And before that one, in this lecture series, you all have studied about autoencoders, variational autoencoders. Then we have studied GANs, and then today we will study about diffusion models. These All these models are generative models that are majorly used for generating images, videos, audios. Okay?

[01:15] Now, what Let us see the time frame between 1980s to till date today. In this time frame, how these evolutions happened. Okay? So, in 1987, autoencoders were introduced. What was the function of the autoencoders? It was used for image

generation. And it was mainly specialized in reconstructing the input from a compressed compressed latent representation. So, you have something like, you know, the example that I can give here is about a file. We have a file or a folder and we zip it, okay? And when we unzip that folder, we get the same things that we have put in the file. So, autoencoders are basically you will take the input, it will compress it into a latent representation, compressed latent

[02:15] representation, and then it reconstructs the same image, okay? From the compressed latent representation. So, as I was telling, they're similar to zip compression. Now, where are the applications for what we were doing? Because we are reconstructing the same image, okay? It was used for removing unwanted noise from data and improve quality. So, noise removal, identifying anomalies in an image and it is also used for extracting important features for better model performances.

[02:54] So, what we see here is there are Uh this is one example where image restoration via you know, autoencoders are used for image restoration. You can see there are some scratches in this image and that has been restored and this has been done using autoencoders. But, what is the limitation of autoencoders? Traditional autoencoders cannot generate realistic new images. So, whatever was there as the input, it is reconstructing that one. So, that is about autoencoders.

[03:29] Then in 2013 variational autoencoders came into picture. Now, what is the specialty of variational autoencoders? They stored the internal representation of an image in terms of a probability distribution. So, instead of you know, a one point, it had a probability distribution associated with the internal representations of an image. And from here, you know, that uh the representation the input image can be represented in terms of the mean and variance of a latent Gaussian distribution, okay? So, that is something which is uh new in variational autoencoders, and the main significant difference autoencoders and variational autoencoders is it can generate new images, okay?

[04:27] So, suppose we have the image of a ship. It is going through the VAE structure, and we see there are various various variations of that same image. So, the image structure is same, it is same ship, but here the color of the sky and the color of the water is changed. Here also it is changed. Here some portion of the uh the ship's uh color has been changed. Here it is white, whereas the input one was black. And, you know, you can see variations here. Here it was white, here it is red. So, these are the variations of the input image. So, that was that became possible in variational autoencoders because the internal representation of the image is kept as a probability distribution. Okay?

[05:16] But, what was its limitation? It often the generated images were blurry. So, they were not very high-resolution images that were coming from the VAE. Then, in

2015, the GANs came into picture. And GANs were able to create high-quality fake images from noise. And, you know, apps like FaceApp in 2017 and this person does not exist used GANs for their um um GANs as their backbone architecture. What was the limitation of GANs?

[05:56] Limitation was one is mode collapse. What do you mean by mode collapse? Mode collapse means often the generated images are very similar to the very similar to each other. So, there is not much of variation in the generated images. So, suppose you have a training set where you have images of dogs, cats, uh horses, and birds. And when the generated images are coming out of the GANs, it is only generating probably dogs images. So, this is called as a mode collapse. And also the training is uh training uh of GANs is a challenge. It is very unstable for training purpose, okay? So, training instability is a major limitation in GANs.

[06:40] Then in 2015, diffusion models came into existence. So, diffusion model, from the name itself we can tell. Diffusion models, these are the models which work on the principle of diffusion. And what is diffusion? Diffusion is a phenomena of physics where particles spread from a region of higher concentration to a region of lower concentration. So, that is called diffusion. So, it is a very basic physics phenomena. And we will see how this phenomena came uh or it is used in generative models like diffusion probabilistic models, okay?

[07:22] So, in 2015, this diffusion models came into picture. And why it came into picture? Because we wanted to two things, or scientists wanted two things. One is the model should be stable while we are training, like VAEs, but the image quality should be like GANs. So, these two things were not coming together, so diffusion models got introduced, okay? Now, we will have a look into what is this diffusion model, how diffusion, which is a physics phenomena, can be related with something or uh you know, became a uh uh you know, significant thing in uh generative models. So, we will have a look into that one. So, in 2015, uh researchers connected machine learning with the concept of non- non-equilibrium

[08:16] thermodynamics, which is a branch of physics which speaks about the movement of particles from a region of higher concentration to a region of lower concentration, okay? So, if we see a picture here, so physics of particle motion and generative AI, these two are bridged by the diffusion model, okay? And if we see the picture, this is a glass of water in which one drop of red ink has been dropped. And we all know when ink is dropped, it starts spreading, okay?

[08:50] So, this process of spreading of ink or uh a smoke or anything, you know, from a region of lower concentration higher concentration to a region of lower concentration, this is the process of diffusion, okay? Now, we'll try to see what exactly or how diffusion came into picture in the generative models. Let us take a coordinate space, XYZ, and we call it as a data landscape. So, I will show or represent data in this

coordinate system or in this landscape. And this coordinate system will be shaped by probability. What I What do I I by that?

[09:35] If I have coordinates X_0 X_1 and then the Z axis is my probability. Okay? Now, we will see how we can visualize data in this landscape. I have drawn a spike here. That means this is a region which is having high probability compared to these regions. And these regions of higher probability, another one is another spike is here, they represent realistic images in the data distribution. So, if we have high probability region, we can we can imagine them as or we can uh match them with the realistic image in data distribution.

[10:27] Okay? Now, there are some spaces where it is uh low probability region. So, that low probability region is basically your noise, which is your unstructured data. So, regions of higher probability correspond to structured data, which is your realistic-looking images. And regions of low probability, that those are your noises, which are unstructured data. So, this is the data landscape. Now, this I have shown in a very very simplified way. In actual scenario, in pixel space, where each pixel will have some numbers, so each pixel channel will have its own axis. So, it will not be like this simple uh you know, three-axis diagram.

[11:12] So, it will have multiple axes, each belonging one one axis belonging to uh each channel of the pixel space. Okay? For simplicity, for making uh you know, visually uh simple that the concept, I have taken only X_0 and X_1 . But, in real practical scenario, this will this can go to a large number depending on the number of channels that are present in that pixel space. Now, how do we visualize data generation? So, generative models basically they generate images.

[11:50] So, it is a process of data generation. Now, when we have looked into this data landscape where high probability regions have given you are correspondent to realistic images and low probability regions correspond to noises, how can I visualize the data generation in that landscape? How can I do that one? So, I have kept the same figure here and if you see, there is if I want to you know, generate a data. Okay, generate a data means here it is an image.

[12:25] If I want to generate an image, so I can start from any point here. And then I should go and go to the top. Okay, top means some high probability region. It can be here, it can be here or some other place. Okay. » [snorts] » Now, the problem is I don't know the landscape. So, the data distribution or the true data distribution is unknown to me. So, I don't know where exactly the peaks are. Where exactly the regions of high probability are.

[12:57] Okay, if I am you know, so how do I navigate my path? I will be lost somewhere, right? And I'll not be able to generate a realistic looking data which is a realistic looking image in this case. So, one solution that we can have is we have all the images in the world and then map them in the data landscape, you get so many

peaks like this, you know, all the images in the But, that is impossible. That is not possible practically, okay? And also, if you have this is something which is your known territory. So, you have whatever known images you had, you have mapped them in this landscape. But, what about an unknown space or unknown image? So, in that case, you will not have any clue how to navigate. Learning to navigate, it is

[13:52] like, you know, you know in your house where are the rooms. You know which which way you should go to find your kitchen, to find your dining space. But, that does not mean that you can navigate smoothly without any hindrance in some other house, okay? So, when such a thing unknown scenario comes, we will not be able to generate new data, okay? So, that is not possible. So, what exactly is needed here? We need a model which can generalize which has the ability to navigate in general, okay? It can navigate anywhere. When I tell navigate anywhere, means it can start from any place and then navigate in a correct direction so that it reaches the top of the or a high probability region.

[14:44] Which corresponds to a realistic data, okay? Now, when I tell it should have uh navigational abilities, general navigational abilities, I mean that at each and every local each and every point here, suppose the point is here, it should be able to judge which direction should I go so that I keep increasing I keep going up the hill, okay? So, what should be my next step? So, that is that is what I mean to say when I'm talk telling that it should have general general navigational abilities. That means suppose it is standing here and then should it go this side or this side or this side? It should be able to make a decision by its own, okay? So, it is like a compass.

[15:42] It will tell when you when you are at point this, you should go in this direction to go to a region of high probability. Okay? So, that is something I want my model to do. So, it should work like a compass. Now how do we do that navigation and how physics comes or diffusion models uh or physics comes into diffusion models. So, we will see. When I'm telling I want my model to navigate the path, that means at each and every point it should see its surrounding and judge which direction it should move so that it has the ability it can go to the high probability region. That is where when if it reaches there, it means it will it has created some realistic looking image, okay?

[16:37] Now this is basically a time varying probability distribution, okay? W- Why do I uh tell it is a time varying probability distribution? Because each and every step it is taking the next step it is taking, it is it is varying with time. So, at the first instance, whatever step it has taken at the second instance of time, it will take some other step. So, it is a time varying probability distribution. So, instead of P_X , we will work with $P_{\text{subscript } T \text{ of } X}$. So, time also comes into picture. And this is where physics comes into picture because we will write it like this. $P_{\text{subscript } t \text{ of } x}$ is equal to p_x given t . That is probability of data x at time t . Okay. So, we are having both the space and the time uh parameters

[17:35] here in this probability distribution. And this is a new concept in machine learning or generative AI. And this is where physics comes into picture where in physics it is very very common to have time dependent fields. So, you can think of the ink spreading in the water. It is time dependent because initially at time equal to t equal to zero the uh when you drop the ink in the water in a glass of water, it is concentrated in one point. At t equal to 5 seconds, it has spread. So, it is a time varying uh uh phenomena.

[18:16] When we are telling the exploration or the model's ability to navigate the path towards the region of higher concentra- higher probability, it is a time varying probability distribu- it is a time varying phenomena. That is where your physics comes into picture. Okay. So, concentration changes over space and time. Both are Both are associated in diffusion. Now, this next question is why were diffusion models introduced?

[18:47] What was the problem of the existing generative models? We have seen uh VAEs. We have seen autoencoders, VAEs, GANs. Why the need came for the diffusion models? Now, most generative models or the existing generative models falls in one of the one of these categories below. They can either be tractable but not uh efficient enough to model complex real world data, okay? Or handle real world data like speech, text, etc.

[19:25] Tractable, when I tell tractable, it means we can compute the probabilities, train the model, optimize the parameters, sample it, or generate data from it. So, the mathematical uh operations are very simple. So, that is meant by tractable. So, either the generative model, whatever existing generative models that we had, they are either tractable but not expressive, that is, they are unable to handle real world complex data like images, speeches, etc. Or they can be expressive expressive, that means they are able to handle real world data but not tractable. So, we cannot evaluate how good they are, train it if we cannot train it efficiently, generate high-quality samples efficiently. So, the existing generative models, what was

[20:17] there at that point of time, they fall either of these in these two categories. Either they can model complex data, but in that case they become you know, intractable, or they are tractable but cannot handle handle complex data. So, the concept of diffusion models got introduced in this paper, deep unsupervised learning using non-equilibrium thermodynamics. So, I urge the participants to go through this paper and uh see into the maths part. A detailed maths is given there.

[20:54] Then we will come into why we had diffusion probability why what is the need for the diffusion probabilistic model. So, as I was telling that there can be uh um, tractable intractable problem, or they can be expressive or non-expressive. Now, we want a model which is both flexible, that is it is able to handle complex data. At the same time, it is computationally tractable. So, both these things we wanted.

And diffusion models solve this by breaking hard problems into many easy steps. So, diffusion models came into picture which gave both of these, that is highly flexible as well as they are computationally tractable.

[21:42] Now, how diffusion models work you know, tackle this problem? In diffusion model, there are two processes. One is called as the forward process which is a known and fixed process. Another one is called as the you know, reverse process, okay? Which is where the learning happens. So, we look into what is the forward process. Forward process basically, you take an image and then you keep on adding noise to it till it becomes complete noise, Gaussian noise, okay? That means you are changing your structured data or complex data in the form of image to a simple Gaussian which is basically your noise. So, it goes like this. If X_0 is your image input, it goes through a number of steps $X_1 X_2 X_3 \dots X_T$ capital

[22:35] and then X_T belongs to a Gaussian distribution. Okay? So, we are changing the complex data to something which is Gaussian and simple to mathematically simple to work with. This is easy and tractable. Whereas in reverse diffusion process or reverse process, that is the place where the learning happens, okay? And what what exactly is done in the reverse process? It will become more clear when we go forward with the the visuals and the maths part. But right now, you understand there is a forward process which takes your input image, starts giving noise to it in each and every step. Little bit, little bit, little bit noises are added till the point where you get a complete Gaussian noise, okay? So, from structured data to

[23:28] completely random unstructured data. And why we have done it? We did it because it is easy and tractable. Okay? And then once you have gone, then the reverse process happens where you learn to reverse noise step by step. So, there in that case you start from the last one, which is X_T , which is the complete noise, and then you keep on going in the reverse direction like X_T minus one, X_T minus two till you go back to X_0 , which is your image data, original image data.

[24:01] And we have now broken it into various steps. The steps number of steps may vary largely. It can go from 100 steps, uh you know, 200 steps depending on the complexity of your problem statement. And because you have started from Gaussian and you are coming towards a structured data, each denoising step becomes easier to learn, okay? Now, we will have a look. Till now we have whatever we have learned, we'll have a look how we can till all the things or bring them in some points for diffusion models. So, diffusion models are basically generative AI models which are used for um generating images, audio, video from noise. And it has a controlled process of adding noise to the image and the model learns

[24:59] once you take your image keep adding noise to it in a number of steps and goes and go finally to a Gaussian noise and then you the model then learns to reverse the denoising process. Okay? So, it starts denoising, that is taking out noise from

the image and it keeps on doing it in through multiple steps till it goes to the original image. Okay? And so, if we have just another visual interpretation or you know, like a flowchart I've given. Suppose you have an image of a human face, then in the forward diffusion it becomes blurry because you have added some noise to that one. Then it becomes more when you add more noise to it it becomes a random pattern and then finally it becomes pure Gaussian noise. Okay?

[25:48] So, everything becomes simple distribution like it's a Gaussian distribution. Okay? Now, we'll come to the core idea of diffusion model. What exactly we are trying to do here? We understood there is a forward diffusion process, there is there is a reverse process. And but what exactly is happening here? Now, learning a probability distribution is hard. But learning probability distribution based on just one previous step is easier. So, learning $P(X_{T-1} | X_T)$ is much easier.

[26:33] So, what what do I mean by that one? When you have the probability distribution of X_T we you know that what is X_T and then from X_T finding out what will be X_{T-1} Okay? That probability distribution of $X_{T-1} | X_T$ is much easier than to find $P(X)$. So, that is the core idea for diffusion models and that is the reason it is broken into a number of steps. So, what we do here is in forward diffusion process, we start from a clear image that is say your X_0 , then X_1, X_2, X_3 and goes to $X_{small\ T}$ and then at T equal to capital T , it is almost pure noise. So, in the forward diffusion process, there is no learning that is happening. It is just corrupting your image data. Okay.

[27:32] In the reverse diffusion process, that is where your model learns. Now, what does the model not learn? The model learns how much the model learns to denoise the image. So, you have started from the pure you know, noise, almost pure noise and then you have taken out some noise from there and in the next step, some more noise is taken out. So, you are denoising the image. At each and every step, you are denoising the image and you continue this one till you go to the clean image.

[28:06] Okay. So, this reverse process is actually the learning process. Okay. Where your model learns that this much of noise is there in the image and it is denoising the image by that learning. Now, if we look at the training and the inference pipeline, it will become much more clearer, I presume. So, in the training pipeline, what happens is we take a real image X_0 , add noise to it to get X_T in a in a process of multiple steps. We break it down into multiple steps. So, X_0 , then X_1, X_2 , it can go to X_{100} . Okay?

[28:51] And finally, we get a noisy image. And this noisy image goes to the neural network. And this neural network predicts the added noise. So, you have an image. Uh suppose I have started from X_0 , then add noise to get X_T . That might not be the final one. So, X_T and then we give this noisy image to the neural network. And then, what is the function of the neural network? It predicts the noise added in X_T compared to X_0 . Computes denoising loss between true and predicted noise. So, the

model learns how much to Uh it finds out the difference between the actual noise added and the uh ground truth. Okay?

[29:49] So, the loss is basically your predicted noise minus the actual noise. Okay? And once you're computing the loss, then you update the weights of your network so that the prediction of the noise added in this image becomes more accurate. Okay? So, this is the training pipeline. In the inference pipeline, inference means in this case of generative AI, inference means we are sampling or generating some new data. Okay? What happens? The generation starts from pure random noise, which is your reversed uh process in diffusion model.

[30:25] So, here X_T belongs to a Gaussian distribution with zero and one uh mean and variance. And then, we the trained model repeatedly denoises. So, what happens? X_T , then X goes to $X_T - 1$. So, X_T to X_T minus one, now the model knows how much uh noise have been added or how much noise to take out to make the image from X_T to X_T minus one. How much noise should be removed from the image to take the image from X_T minus one to X_T ? And this process continues till X zero.

[31:03] And this repeated denoising gradually generates a realistic-looking image. So, the learning happens in the training phase and in the inference or sampling pipeline, it starts from the noise noise and the trained model keeps denoising that image and it gives repeated denoising gradually generates the realistic-looking image. So, the I've come almost to the end of today's discussion. I will conclude with a little introduction to Markov chain. So, Mark What is Markov chain? Uh Markov chain is uh it describes a system uh that transitions between different situations called states. So, there will be state one, state two, and there will be transition between state one to two.

[31:52] And it can continue to till state state N . So, zero to one, one to two, two to three, and then it continues till N its state. Now, the probability of moving to the next steps depend on the current state and not on the sequence of the states that have been uh you know, that came before it. So, suppose you are going from N minus one its state to the N its state. So, the probability of moving to the N minus one to the N its state will depend only on N minus one its state condition, not on from zero to N minus two states, okay?

[32:28] So, that is Here it is I have explained it with a visual representation having the snake and ladder board game. So, suppose you are in this box nine. So, if you are at nine, then you have the ladder which goes to 55. Similarly, if you are in seven Sorry, if you are at say 98, then you come down to 76. So, looking at this nine, does it matter? Because when you when you are at nine, then only you get the chance to go to 55. It doesn't matter how you have come till nine, whether you have gone up, down, what whatever you have done, it there is absolutely no memory for that one. And it only depends so whether you go to 55 or not depends on whether you are in

nine. Okay? If you are at eight, you are not directly going

[33:22] to 55 that that block. So, this is what is Markov chain. Mathematically, this is expressed as to generate X_T only X_{T-1} is required. Okay? So, that is This is a mathematical representation for Markov chain. And why we are introducing Markov chain here? Because both forward noising process and reverse denoising process are modeled as Markov chain. So, with this one, I conclude today's session. Before I conclude, let us go and have a quick look on the summary.

[34:02] And in the summary, we would like to take you through what was the reason for diffusion models coming into picture, what was the necessity? Because we have seen VAEs, autoencoders, variational autoencoders, and GANs, they have given good quality images, but there was either problem on the training stability or on the quality of the images. We wanted both of them together, and that This reason from where diffusion models came into picture, where they are able to produce images from noise. And we have explained how the data distribution landscape looks like in the probability driven data distribution landscape. How it looks, how we can visualize that one.

[34:48] And how if I want to navigate and you know, if I want to generate a new image, how should I navigate? What are the What are the things that I would look into my look inside my model so that it can have a generalized navigation ability. Okay. So that that thing we have looked into. And we have seen that if we break down this process as time and space varying process, which is dependent on time and space both, it becomes easier for the model to navigate because when it is navigating from one point to the other in the data distribution landscape, it the space and the time both are varying.

[35:34] Okay. So it is a time and space varying domain. And that is where physics comes into picture because in physics it is very common to have very common to have fields which are space and time varying. Okay. So from this perspective, this diffusion name came into picture. And in diffusion models we have two processes, the forward reverse forward noising process and the reverse denoising process. In the forward noising process, we take an image and corrupt it with small small little amount of noise at each and every step still we get a Gaussian noise. In the reverse denoising process, we take out noise from an image and make a clearer images. So multiple steps of denoising will give us a realistic looking image.

[36:25] So we have looked into the training pipeline and the inference pipeline also, and we have finished with a little touch about the Markov chain, which tells that uh the probability of a a particle to be at any instant depends only on the previous instance of time, okay? So, it has nothing to do with the all the states that it has gone. Now, the probability of a particle to be in a state uh S depends only on the previous step, not on all the previous step, just to be before that one. Whatever is the current

state that time, that will decide what will be the probability of going to another state. So, with this one, I conclude today's session. In the next uh lecture session, we will discuss about

[37:07] the mathematics that is behind diffusion models. And uh till that time, till I meet you all, uh continue learning. Happy learning. Thank you all.